

Digital logic

- One of the most important branch of Electronic and telecommunication Science sector is Digital electronics(logic). Digital logic is mainly used for data(must be digital information) representation, manipulation and processing of using discrete signals or binary digits (bits). It can perform logical operations, data retrieval or storing and data transformation by analyzing logical circuit design.

Why Digital Logic is Necessary?

- In modern computing realm, Digital logic plays a significant role in many sectors which are discussed below :
- **Universal Representation:** For any type of data representation like image, text, video, audio etc. digital logic/system is used by encoding the data in binary form. This binary formatted data enables uniform handling of diverse data and allows seamless integration and compatibility.

Why Digital Logic is Necessary?

- **Error Reduction and Correction:** Digital logic itself is very less prone to error as it works with only two values(0 and 1). Moreover, we can employ redundancy check and error detection mechanisms by digital logic codes which can detect and rectify errors introduced during transmission. This ensures reliable and accurate data processing.
- **Scalability and Modularity:** Digital logic provides scalable framework by which we can develop complex system by using basic logic gates only. This enables a easy and cost effective way to develop a large-scale system with improved flexibility, maintainability, and ease of integration.

Why Digital Logic is Necessary?

- **Noise Immunity:** As digital logic follows the discrete nature of signal so it is less prone to have induced noise compared to analog signal. So it provides more robust communication and data processing by noise filtering and error mitigation.

Analog signal

- Analog signal is a type of signal that represents continuous data using a continuous range of values. In other words, it can take on any value within a certain range. Analog signals are characterized by their smooth and continuous nature. For example, imagine a sound wave. In analog form, the sound wave is represented by a continuously varying electrical signal that mirrors the fluctuations in air pressure caused by the sound.

Digital signal

- Digital signal is a type of signal that represents data as a sequence of discrete values, typically using binary numbers, it also contain different voltage values. Unlike analog signals, which vary continuously over time, digital signals are discrete and quantized, meaning they only take on specific, distinct values.
- In digital communication and computing systems, information is encoded into digital signals for transmission, processing, and storage. These signals are commonly used in telecommunications, audio and video processing, computer networks, and many other applications.

Basis	Analog Signal	Digital Signal
Definition	Analog signals represent continuous variations in magnitude over time.	Digital signals are Discrete and quantized, with specific values.
Signal Type	Continuous waveforms	Discrete Signals
Processing	Requires complex processing for manipulation.	Easier to process and manipulate digitally.
Storage	Less efficient for storage due to continuous nature.	More efficient for storage due to discrete values.
Bandwidth	Typically requires more bandwidth.	Requires less bandwidth for transmission.
Examples	Analog audio signals, analog radio waves, Human voice, etc.	Digital audio signals, digital data streams, computers, etc.
Errors	Susceptible to noise and distortion	More resistant to noise and distortion
Circuit Component	Amplifiers, filters, continuous-wave oscillators	Microprocessors, binary counters, logic gates
Signal Values	Infinite range of values	Limited to discrete values
Conversion	No conversion required	Analog-to-digital conversion (ADC) required
Applications	Analog signals are used in electric fan, landlines, radio frequency communications, etc.	Digital signals are used in computers, smartphones, digital sensors, digital imaging, etc.